

William J. Oldroyd

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TECHNICAL/RESEARCH POSITIONS

Sr. SSA Solutions Engineer LeoLabs	Sep 2026 – Present Remote
Sr. Astrodynamics Engineer, SDA Katalyst Space Technologies	Jul 2025 – Sep 2026 Remote
Research Fellow Katalyst Space Technologies	Jun 2024 – Jun 2025 Remote
Postdoctoral Research Scholar/Researcher Northern Arizona University	Aug 2022 – Present Remote

EDUCATION

Northern Arizona University (NAU) Astronomy and Planetary Science, MS, PhD	Aug 2018 – Aug 2022 Flagstaff, AZ
Brigham Young University (BYU) Physics-Astronomy, BS; Minors: Mathematics, Spanish, Geology	Aug 2010 – Dec 2011 & Apr 2014 – Apr 2018 Provo, UT

SELECT PUBLICATIONS

Refereed (20)

- *NSF-DOE Vera C. Rubin Observatory Observations of Interstellar Comet 3I/ATLAS (C/2025 N1)* - Chandler, C., et al. including Oldroyd, W., ApJL, 1001, L35, Apr 2026
- *An Efficient Shift-and-stack Algorithm Applied to Detection Catalogs* - Stetzler, S., et al. including Oldroyd, W., AJ, 170, 352, Dec 2025
- *Recurrent Cometary Activity Discovered on Quasi-Hilda Jupiter Family Comet 362P/(457175) 2008 GO98* - Farrell, K., Chandler, C., Trujillo, C., Oldroyd, W., et al., ApJL, 987, L35, Jul 2025
- *A Dormant Captured Oort Cloud Comet Awakens: (18916) 2000 OG44* - Chandler, C., Oldroyd, W., et al., ApJL, 986, L2, Jun 2025
- *The Volatile Composition and Activity Evolution of Main-Belt Comet 358P/PANSTARRS* - Hsieh, H., et al. including Oldroyd, W., PSJ, 6, 3, Jan 2025
- *The DECam Ecliptic Exploration Project (DEEP) VII: The Strengths of Three Super-Fast Rotating Main-Belt Asteroids from a Preliminary Search of DEEP Data* - Strauss, R., et al. including Oldroyd, W., AJ, 168, 184, Oct 2024
- *The Active Asteroids Citizen Science Program: Overview and First Results* - Chandler, C., Trujillo, C., Oldroyd, W., et al., AJ, 167, 156, Apr 2024
- *Beyond Point Masses III: Detecting Haumea's Nonspherical Gravitational Field* - Proudfoot, B., et al. including Oldroyd, W., PSJ, 5, 69, Mar 2024
- *The DECam Ecliptic Exploration Project (DEEP) VI: first multi-year observations of trans-Neptunian objects* - Smotherman, H., et al. including Oldroyd, W., AJ, 167, 136, Mar 2024
- *The DECam Ecliptic Exploration Project (DEEP) IV: Constraints on the shape distribution of bright TNOs* - Strauss, R., et al. including Oldroyd, W., AJ, 167, 135, Mar 2024
- *The DECam Ecliptic Exploration Project (DEEP) III: Survey characterization and simulation methods* -

Bernardinelli, P., et al. including Oldroyd, W., AJ, 167, 134, Mar 2024

- ***The DECam Ecliptic Exploration Project (DEEP) II. Observational Strategy and Design*** - Trujillo, C., et al. including Oldroyd, W., AJ, 167, 133, Mar 2024
- ***The DECam Ecliptic Exploration Project (DEEP): I. Survey description, science questions, and technical demonstration*** - Trilling, D., et al. including Oldroyd, W., AJ, 167, 132, Mar 2024
- ***The DECam Ecliptic Exploration Project (DEEP): V. The Absolute Magnitude Distribution of the Cold Classical Kuiper Belt*** - Napier, K., et al. including Oldroyd, W., PSJ, 5, 50, Feb 2024
- ***Recurring Activity Discovered on Quasi-Hilda 2009 DQ118*** - Oldroyd, W., et al., ApJL, 957, L1, Nov 2023
- ***Tuning the Legacy Survey of Space and Time (LSST) Observing Strategy for Solar System Science*** - Schwamb, M., et al. including Oldroyd, W., ApJSS, 266, 22, Jun 2023
- ***Migratory Outbursting Quasi-Hilda Object 282P/(323137) 2003 BM80*** - Chandler, C., Oldroyd, W., & Trujillo, C., ApJL, 937, L2, Sep 2022
- ***Photometric observations of the binary near-Earth asteroid (65803) Didymos in 2015-2021 in support of the DART space mission*** - Pravec, P., et al. including Oldroyd, W., PSJ, 3, 175, Jul 2022
- ***Outer Solar System Perihelion Gap Formation Through Interactions with a Hypothetical Distant Giant Planet*** - Oldroyd, W., & Trujillo, C., AJ, 162, 39, Aug 2021
- ***Cometary Activity Discovered on a Distant Centaur: A Non-Aqueous Sublimation Mechanism*** - Chandler, C., Kueny, J., Trujillo, C., Trilling, D., & Oldroyd, W., ApJL, 892, L38, Apr 2020

Research Notes and White Papers (25)

- ***The 2023 Reactivation of Main-belt Comet 427P/ATLAS*** - Hsieh, H., Thirouin, A., Pittichová, J., Chandler, C., Oldroyd, W., RNAAS, 10, 267, Sep 2026
- ***A Tail of Two Classifications: Citizen Science Reveals Activity from Jupiter-family Comet 2021 RK133*** - Singh, D., et al. including Oldroyd, W., RNAAS, 10, 210, Jul 2026
- ***The Mars-crossing Asteroid that was Actually a Comet: Citizen Science and AI Reveal Activity from (446816) 2000 EE38*** - Morato, N., Chandler, C., Sedaghat, N., Oldroyd, W., et al., RNAAS, 10, 207, Jul 2026
- ***Activity Discovered on Jupiter Family Comet 2015 DD349*** - Farrell, K., et al. including Oldroyd, W., RNAAS, 10, 163, Jun 2026
- ***AI-enhanced Citizen Science Discovers Cometary Activity on Near-Earth Object (523822) 2012 DG61*** - Chandler, C., Sedaghat, N., Oldroyd, W., et al., RNAAS, 9, 3, Jan 2025
- ***Citizen Scientists Discover "Asteroid in a Cometary Orbit" 2010 MK43 has a Comet Tail*** - Kueny, J., Chandler, C., Frissell, M. K., Oldroyd, W., et al., RNAAS, 8, 235, Sep 2024
- ***Cometary Activity from Minor Planet 2015 VP51 Discovered with Citizen Science*** - Frissell, M. K., Chandler, C., Oldroyd, W., et al., RNAAS, 8, 225, Sep 2024
- ***Discovery of Jupiter Family Comet 2011 UG104 Through AI Enhanced Citizen Science*** - DeSpain, J., Chandler, C., Sedaghat, N., Oldroyd, W., et al., RNAAS, 8, 140, May 2024
- ***2016 UU121: An Active Asteroid Discovery via AI-enhanced Citizen Science*** - Sedaghat, N., Chandler, C., Oldroyd, W., et al., RNAAS, 8, 51, Feb 2024
- ***AI-enhanced Citizen Science Discovery of an Active Asteroid: (410590) 2008 GB140*** - Chandler, C., Sedaghat, N., Oldroyd, W., et al., RNAAS, 8, 50, Feb 2024
- ***Activity Discovered on Mars-Crossing Jupiter Family Comet 2018 OR by Citizen Scientists*** - Farrell, K., Chandler, C., Oldroyd, W., et al., RNAAS, 8, 18, Jan 2024
- ***Cometary Activity Discovered on Vacationing Centaur 2019 OE31*** - Oldroyd, W., et al., RNAAS, 7, 270, Dec 2023
- ***Mars-Crossing Minor Planet 2018 VL10: a Jupiter-family Comet Discovery via Citizen Science*** -

Chandler, C., Oldroyd, W., et al., RNAAS, 7 279, Dec 2023

- ***New Active Jupiter Family Comet 2008 QZ44: a Discovery with Citizen Science*** - Chandler, C., Oldroyd, W., et al., RNAAS, 7 271, Dec 2023
- ***Citizen Science Discovery of a Recurrently Active Jupiter-Family Comet: (551023) 2012 UQ192*** - DeSpain, J., Chandler, C., Oldroyd, W., et al., RNAAS, 7, 257, Dec 2023
- ***New Active Quasi-Hilda Asteroid 2004 CV50: A Citizen Science Discovery*** - Chandler, C., Oldroyd, W., et al., RNAAS, 7, 237, Nov 2023
- ***A Jupiter-family Comet Discovery via Citizen Science: 2005 XR132*** - Chandler, C., Oldroyd, W., et al., RNAAS, 7,170, Aug 2023
- ***Cometary Activity on Quasi-Hilda Object 2018 CZ16*** - Trujillo, C., Chandler, C., Oldroyd, W., et al., RNAAS, 7,106, May 2023
- ***New Active Asteroid (588045) 2007 FZ18*** - Chandler, C., Oldroyd, W., et al., RNAAS, 7, 102, May 2023
- ***New Recurrently Active Main-belt Comet 2010 LH15*** - Chandler, C., Oldroyd, W., et al., RNAAS, 7, 60, Mar 2023
- ***Comet-like Activity Discovered on Quasi-Hilda Asteroid 2009 DQ118*** - Oldroyd, W., et al., RNAAS, 7, 42, Mar 2023
- ***New Active Asteroid 2015 VA108: A Citizen Science Discovery*** - Chandler, C., Oldroyd, W., et al., RNAAS, 7, 27, Feb 2023
- ***Discovery of Dust Emission Activity Emanating from Main-belt Asteroid 2015 FW412*** - Chandler, C., Trujillo, C., Oldroyd, W., et al., RNAAS, 7, 22, Feb 2023
- ***Community Challenges in the Era of Petabyte-Scale Sky Surveys*** - Kelley, M., et al. including Oldroyd, W., BAAS, 53, 495, May 2021
- ***The Scientific Impact of the Vera C. Rubin Observatory's Legacy Survey of Space and Time (LSST) for Solar System Science*** - Vera C. Rubin Observatory LSST Solar System Science Collaboration et al. including Oldroyd, W., BAAS, 53, 236, May 2021

Select Recent Conference Presentations (7 of 61)

- ***Adaptive Cislunar Sensor Tasking Optimization through Constellation Design and Admissible Region IOD*** - Oldroyd, W., Uyekawa, J., Lewis, A., Howell, K., Frueh, C., CLSC6, Laurel, MD, Dec 2025
- ***Dynamic Sensor Tasking for Automated Resident Space Object Characterization*** - Oldroyd, W., Uyekawa, J., Correa, J., Aguirre, F., AMOS26, Maui, HI, Sep 2025
- ***Simultaneous Orbital and Physical Property Estimation for Space Domain Awareness Using Augmented State Estimation Filtering*** - Correa, J., Oldroyd, W., Uyekawa, J., AMOS26, Maui, HI, Sep 2025
- ***Enhanced Non-Kinematic Observable Utilization for UCT Correlation*** - Oldroyd, W., Ware, J., & Jones, G., SFM35, Kaua'i, HI, Jan 2025
- ***Orbital and Morphological Evolution of Twin-Tailed Comet 478P*** - Oldroyd, W., Chandler, C., Trujillo, C., & Deen, S., DPS56, 213.06, Boise, ID, Oct 2024
- ***Post-Maneuver UCT Correlation Using Multi-Source Data Streams*** - Hofer, G., Jones, G., Oldroyd, W., AMOS25, id44, Maui, HI, Sep 2024
- ***Dynamical Migration of Discoveries from the Active Asteroids Citizen Science Project*** - Oldroyd, W., Chandler, C., & Trujillo, C., DDA55, 405.29, Toronto, Canada, May 2024

Minor Planet Discoveries (155)

- | | | |
|--|-------------------------------|-------------------------------|
| • 2015 HP198 = 2019 GW183 - MPEC 2024-K155 | • 2021 TK192 - MPEC 2023-X31 | • 2021 RS237 - MPEC 2023-W158 |
| • 2021 CG56 - MPEC 2024-G25 | • 2021 TJ192 - MPEC 2023-X30 | • 2020 UJ74 - MPEC 2023-W158 |
| • 2020 YP30 - MPEC 2024-G23 | • 2021 TH192 - MPEC 2023-X29 | • 2020 UH74 - MPEC 2023-W158 |
| • 2020 UT74 - MPEC 2023-X36 | • 2019 QD112 - MPEC 2023-X28 | • 2020 UG74 - MPEC 2023-W158 |
| • 2021 RN238 - MPEC 2023-X35 | • 2021 RX237 - MPEC 2023-W158 | • 2020 UF74 - MPEC 2023-W158 |
| • 2021 RM238 - MPEC 2023-X34 | • 2021 RW237 - MPEC 2023-W158 | • 2019 QT111 - MPEC 2023-W158 |
| • 2019 QF112 - MPEC 2023-X33 | • 2021 RV237 - MPEC 2023-W158 | • 2019 QS111 - MPEC 2023-W158 |
| • 2019 QE112 - MPEC 2023-X32 | • 2021 RU237 - MPEC 2023-W158 | • 2021 RR237 - MPEC 2023-W157 |
| | • 2021 RT237 - MPEC 2023-W158 | • 2020 UE74 - MPEC 2023-W157 |

- 2020 UD74 - MPEC 2023-W157
- 2020 UC74 - MPEC 2023-W157
- 2019 QR111 - MPEC 2023-W157
- 2019 QQ111 - MPEC 2023-W157
- 2019 QP111 - MPEC 2023-W157
- 2019 QO111 - MPEC 2023-W157
- 2019 QN111 - MPEC 2023-W157
- 2019 QM111 - MPEC 2023-W157
- 2019 QL111 - MPEC 2023-W157
- 2021 RN237 - MPEC 2023-W155
- 2021 RM237 - MPEC 2023-W155
- 2021 RL237 - MPEC 2023-W155
- 2020 UA74 - MPEC 2023-W155
- 2019 QJ111 - MPEC 2023-W155
- 2019 QH111 - MPEC 2023-W155
- 2019 QG111 - MPEC 2023-W155
- 2019 QF111 - MPEC 2023-W155
- 2021 RK237 - MPEC 2023-W154
- 2021 RJ237 - MPEC 2023-W153
- 2019 SP235 - MPEC 2023-W153
- 2019 SO235 - MPEC 2023-W153
- 2019 QE111 - MPEC 2023-W153
- 2019 QD111 - MPEC 2023-W153
- 2019 QC111 - MPEC 2023-W153
- 2019 QB111 - MPEC 2023-W153
- 2019 QA111 - MPEC 2023-W153
- 2019 QZ110 - MPEC 2023-W153
- 2021 RH237 - MPEC 2023-W152
- 2021 RG237 - MPEC 2023-W152
- 2021 RF237 - MPEC 2023-W152
- 2019 QY110 - MPEC 2023-W152
- 2019 QX110 - MPEC 2023-W152
- 2019 QW110 - MPEC 2023-W152
- 2019 QV110 - MPEC 2023-W152
- 2021 RE237 - MPEC 2023-W149
- 2021 RD237 - MPEC 2023-W149
- 2021 RC237 - MPEC 2023-W149
- 2019 SN235 - MPEC 2023-W149
- 2019 QU110 - MPEC 2023-W149
- 2019 QT110 - MPEC 2023-W149
- 2019 QS110 - MPEC 2023-W149
- 2019 QR110 - MPEC 2023-W149
- 2021 TP191 - MPEC 2023-W148
- 2021 RB237 - MPEC 2023-W148
- 2021 RA237 - MPEC 2023-W148
- 2019 SM235 - MPEC 2023-W148
- 2019 QQ110 - MPEC 2023-W148
- 2019 QP110 - MPEC 2023-W148
- 2019 QO110 - MPEC 2023-W148
- 2019 QN110 - MPEC 2023-W148
- 2019 QM110 - MPEC 2023-W147
- 2019 QL110 - MPEC 2023-W147
- 2020 UZ73 - MPEC 2023-W146
- 2021 RZ236 - MPEC 2023-W145
- 2021 RY236 - MPEC 2023-W145
- 2020 UY73 - MPEC 2023-W145
- 2019 SL235 - MPEC 2023-W145
- 2019 SK235 - MPEC 2023-W145
- 2019 QK110 - MPEC 2023-W145
- 2019 QJ110 - MPEC 2023-W145
- 2019 QH110 - MPEC 2023-W145
- 2019 QG110 - MPEC 2023-W141
- 2019 QF110 - MPEC 2023-W140
- 2020 UX73 - MPEC 2023-W139
- 2019 SJ235 - MPEC 2023-W139
- 2019 SH235 - MPEC 2023-W139
- 2019 QE110 - MPEC 2023-W139
- 2019 QD110 - MPEC 2023-W139
- 2019 QC110 - MPEC 2023-W139
- 2019 QB110 - MPEC 2023-W139
- 2019 QA110 - MPEC 2023-W139
- 2019 QZ109 - MPEC 2023-W139
- 2019 QY109 - MPEC 2023-W139
- 2019 SB235 - MPEC 2023-W137
- 2021 RX236 - MPEC 2023-W136
- 2021 RW236 - MPEC 2023-W136
- 2021 RV236 - MPEC 2023-W136
- 2021 RU236 - MPEC 2023-W136
- 2019 QV109 - MPEC 2023-W136
- 2019 QU109 - MPEC 2023-W136
- 2019 QT109 - MPEC 2023-W136
- 2019 QS109 - MPEC 2023-W136
- 2021 LO43 - MPEC 2023-C118
- 2021 FQ54 - MPEC 2023-C116
- 2021 LM43 - MPEC 2023-C114
- 2021 DM19 - MPEC 2023-C113
- 2021 RV209 - MPEC 2023-C112
- 2021 DK18 - MPEC 2022-S119
- 2021 RR205 - MPEC 2022-S118
- 2021 DH17 - MPEC 2022-K183
- 2021 CP40 - MPEC 2022-K182
- 2020 BF102 - MPEC 2022-K181
- 2021 DG17 - MPEC 2022-K180
- 2020 FR40 - MPEC 2022-K179
- 2020 FQ40 - MPEC 2022-K178
- 2020 FP40 - MPEC 2022-K174
- 2021 LL37 - MPEC 2022-K173
- 2020 BE102 - MPEC 2022-K172
- 2021 DF17 - MPEC 2022-J106
- 2020 BY101 - MPEC 2022-G191
- 2015 AH281 - MPEC 2021-Y32
- 2021 DS15 - MPEC 2021-Y30
- 2020 BC95 - MPEC 2021-Y29
- 2021 DR15 - MPEC 2021-Y28
- 2021 DO15 - MPEC 2021-Y25
- 2019 GZ129 - MPEC 2021-Y24
- 2019 EV5 - MPEC 2021-Y23
- 2020 VL26 - MPEC 2021-Y21
- 2020 BA95 - MPEC 2021-Y20
- 2019 EU5 - MPEC 2021-Y19
- 2019 ET5 - MPEC 2021-Y18
- 2021 DN15 - MPEC 2021-Y17
- 2019 SH187 - MPEC 2021-Y16
- 2021 DM15 - MPEC 2021-Y09
- 2020 FQ38 - MPEC 2021-Y08
- 2020 BZ94 - MPEC 2021-Y06
- 2019 SS149 - MPEC 2021-D88
- 2020 BR60 - MPEC 2021-D85
- 2019 SW148 - MPEC 2021-C298
- 2020 FB31 - MPEC 2021-C297
- 2020 FA31 - MPEC 2021-C289
- 2020 FY30 - MPEC 2021-C282
- 2015 VX184 - MPEC 2020-L18
- 2017 VO34 - MPEC 2019-F101
- 2018 AX18 - MPEC 2019-C98
- 2017 WH30 - MPEC 2019-C97
- 2017 SN132 - MPEC 2019-C96
- 2017 OG69 - MPEC 2019-C95
- 2018 VG18 - MPEC 2018-Y14

TELESCOPE OBSERVING EXPERIENCE (Over 1,000 Hours)

- Large Binocular Cameras (Red and Blue), 2x8.4m Large Binocular Telescope, Large Binocular Telescope (LBT) Observatory, Arizona, USA, remote queue
- Gemini Multi-Object Spectrograph Imager, 8.1m Gemini South Telescope, Gemini Observatory, Chile, remote queue
- Inamori Magellan Areal Camera and Spectrograph, 6.5m Walter Baade Magellan Telescope, Las Campanas Observatory, Chile, on site and remote
- Large Monolithic Imager, 4.3m Lowell Discovery Telescope (LDT), Lowell Observatory, Arizona, USA, on site and remote
- Dark Energy Camera (DECam), 4m Víctor Blanco Telescope, Cerro Tololo Inter-American Observatory, Chile, remote
- Astrophysical Research Consortium Telescope Imaging Camera (ARCTIC), 3.5m Apache Point

- Observatory (APO), New Mexico, USA, remote
- Vatt4k, 1.8m Vatican Advanced Technology Telescope (VATT), Mount Graham International Observatory, Arizona, USA, on site observer and trained telescope operator
- 0.5m Barry Lutz Telescope, Atmospheric Research Observatory, Northern Arizona University, Arizona, USA, on site
- 0.4m David Derrick Telescope, Orson Pratt Observatory, Brigham Young University, Utah, USA, on site

HIGH PERFORMANCE/CLOUD COMPUTING AND SOFTWARE DEVELOPMENT EXPERIENCE

- Developed, tested, and deployed automated CI/CD SaaS AWS cloud production pipelines for satellite characterization and anomaly detection
- Utilized over 2 million compute hours on the NAU High Performance Computing Cluster, *Monsoon*, for orbital dynamics simulations, space weathering models, exoplanet lightcurve extraction, and parallel computing metric testing
- Programming languages used in research and product development: Python, C/C++, bash, LaTeX, Terraform, Javascript, IDL, IRAF, MATLAB, VBA, Mathematica
- Other software tools used in research/product development: AWS dev tools, Docker, Claude Code, Gitlab, Grafana, Davinci Resolve, Adobe Premiere, Jira, Confluence, Trello, Azure DevOps, Linear, Excel, Word, Powerpoint

TEACHING EXPERIENCE

Metavisionaries Teaching

- Faculty Member and Instructor Jul 2023 – Sep 2023
 - Taught Astrobiology courses through Virtual Reality to college students from a worldwide audience

NAU Mentoring

- Interns to Scholars Mentor Jul 2020 – May 2022
 - Co-mentored a beginning undergraduate in a hands on computational research project focused on searching for exoplanets in data from NASA's Transiting Exoplanet Survey Satellite mission
- Summer Research Experience for Undergraduates (REU) Mentor Jun 2020 – Aug 2020
 - Co-mentored an upper level undergraduate on Constraining the Space Weathering Rate with Asteroid Family Spectral Slope Modeling using NAU's HPC cluster

TRiO Upward Bound Teaching

- Summer Residential Program Instructor Jul 2020, 2021, & 2022
 - Developed curriculum and taught courses on Solar System Exploration and Planet Detection Techniques to high school students from diverse backgrounds, cultures, and academic levels
- South Mountain Community College Saturday Academy Instructor Jan – May 2021 & 2022
 - Developed curriculum and taught courses in Applied Physics and The Science of Natural Disasters to high school students from diverse backgrounds, cultures, and academic levels who are potential first generation college students

BYU Teaching

- Teaching Assistant Dec 2014 – Dec 2017
 - Assisted students in walk-in laboratory with 100-500 level physics labs
 - Repaired, improved and installed equipment for all physics labs in the department
 - Taught Physics 107, a non-physics major lab course

- Support TA over data reduction and analysis using IRAF and AstroImageJ for Physics 329, an upper level observational astronomy course

Tutoring

- Private High School Physics Tutor Sep 2016 – Dec 2016
 - Assisted a student with conceptual understanding and homework

SELECT SCHOLARSHIPS, GRANTS, HONORS, AND AWARDS

- Asteroid 43501 Oldroyd (2001 CW22)
- Space Force Small Business Innovation Research Grant (SBIR) Phase II 2025
- Air Force Small Business Technology Transfer (STTR) Phase II Research Grant 2024
- National Science Foundation Astronomy and Astrophysics Research Grant 2024
- NAU Department of Astronomy and Planetary Science Representative to meet with the US VP Oct 2023
- NAU Graduate Student Government Conference Presentation Competition, Best Presentation 2021
- NAU 3 Minute Research Presentation Competition Peoples' Choice Award 2019
- BYU Office of Research and Creative Activities Student Research Grant 2016-2017

RECENT COLLOQUIA AND OUTREACH PRESENTATIONS

- Discovery Plus Academy STEM Week, 17 Feb 2026, Pima, AZ, “Robots in Space - DART, SIGHT, and LINK”
- Brigham Young University Physics and Astronomy Colloquium, 27 Feb 2025, Provo, UT, “The Search for Planet X”
- Brigham Young University Astronomy Research Group Guest Lecture, 15 Oct 2024, Provo, UT, “Choose Your Own Adventure”
- Discovery Park Public Lecture Series, 24 Aug 2024, Safford, AZ, “The NASA DART Mission”
- Fort Worth Astronomical Society, 18 Jun 2024, Fort Worth, TX, “For the Dinosaurs! Planetary Defense and Comet Formation with NASA's DART Mission”
- Discovery Park Public Lecture Series, 8 Jun 2024, “The Search for Planet X”
- Eastern Arizona College Astronomy 103 Guest Lecture, 22 Apr 2024, Thatcher, AZ, “DART: The Double Asteroid Redirection Test”

ASTRONOMY SERVICE

- American Astronomical Society Division on Dynamical Astronomy Scientific Organizing Committee (SOC) member Sep 2025-Jun 2026
- American Astronomical Society Division on Dynamical Astronomy Equity, Diversity, and Inclusion Year-round Mentoring Program Committee Member and Mentor Jul 2021-Present
- American Astronomical Society Division of Planetary Science-European Planetary Science Congress Meeting Student Mentor Oct 2023
- Referee for Monthly Notices of the Royal Astronomical Society Sep 2023
- Legacy Survey of Space and Time Solar System Science Collaboration Solar System Readiness Sprint Conference Local Host Jun 2020 & Jun 2023
- Lunar and Planetary Science Conference Dwornik Student Presentation Award Judge at 2023 Meeting
- American Astronomical Society Division of Planetary Science meeting Session Chair 2022
- NAU Astronomy and Planetary Science Graduate Student Faculty Representative Jul 2021 - Aug 2022
- American Astronomical Society Division on Dynamical Astronomy meeting Session Chair 2022

- American Astronomical Society Division of Planetary Science meeting Science Chat Moderator, 2020 and 2021
- American Astronomical Society Chambliss Astronomy Achievement Student Awards Poster Competition Judge at the AAS Winter 2021 meeting
- NAU Astronomy and Planetary Science Fellowship Mock Review Panel Panelist Jan 2019, 2020, and 2021

EXTRACURRICULAR, AND VOLUNTEER EXPERIENCE

- Eastern Arizona College Community Bands and Orchestras Jul 2023 – Present
 - Trombone/Bass Trombone Player
- Pima Youth Soccer League Coach Oct 2023 – Present
 - Head Coach at the Junior & Senior levels
- NAU Department of Astronomy and Planetary Science Grad Student Club Oct 2020 – Dec 2021
 - Vice President - coordinated events, organization, and club founding
- Volunteer Boy Scout Leader Dec 2004 – Dec 2018
 - Eagle Scout
 - Merit Badge Counselor for Astronomy and Chess
- BYU Astronomy Research Group Leader Sep 2016 – Dec 2016
 - Planned and oversaw weekly department wide meetings and invited guest speakers
- Volunteer Representative for the Church of Jesus Christ of Latter-day Saints Mar 2012 – Mar 2014
 - Spanish speaking Missionary in Wisconsin
 - Taught English classes for Hispanic minority groups
 - Taught workshops on leadership, goal setting, and teaching
 - Volunteered at hospitals and food pantries
- BYU Student Association Chess Club President Jan 2011 – Dec 2011
 - Organized and competed in intercollegiate chess tournaments, outreach, and weekly meetings